

Engine Only Operates in Rated Mode

Symptoms

- The engine will **only** operate in rated mode.

How To Use This Tree

This symptom tree can be used to troubleshoot engine symptoms. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check Customer Interface Box Wiring	
	STEP 1A. Check Idle/Rated Switch Signal Wire with Engine Harness Disconnected	Less than 10 ohms resistance?
	STEP 1B. Check Idle/Rated Switch Signal Wire with Engine Harness Disconnected	Less than 10 ohms resistance?
STEP 2.	Check Engine Harness to Customer Interface Box Cable	
	STEP 2A. Check Idle/Rated Switch Signal Wire	Less than 10 ohms resistance?

STEP 1. Check Customer Interface Box Wiring

STEP 1A. Check Idle/Rated Switch Signal Wire with Engine Harness Disconnected

Conditions :

- Open the customer interface box
- Disconnect customer interface box to engine harness cable connector C3 from the customer interface box.

Action	Specification/Repair	Next Step
Check idle/rated switch signal wire. • Disconnect the idle/rated switch signal wire from the X4 connector. • Place one test lead on the idle/rated switch signal pin in connector C3. • Place the other test lead on idle rated switch signal pin on the X4 connector.	Less than 10 ohms resistance? YES	1B
	Less than 10 ohms resistance? NO Repair : Replace the wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/115/115-015-023.html).	Repair complete.

STEP 1B. Check Idle/Rated Switch Signal Wire with Engine Harness Disconnected**Conditions :**

- Open the customer interface box.

Action	Specification/Repair	Next Step
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Check idle/rated switch return wire. • Disconnect the idle/rated switch return wire from the X4 connector. • Place one test lead on the idle/rated switch return terminal on the customer interface box logic unit. • Place the other test lead on the idle rated switch return pin on the X4 connector.	Less than 10 ohms resistance? YES	2A
	Less than 10 ohms resistance? NO Repair : Replace the wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/115/115-015-023.html).	Repair complete.

STEP 2. Check Engine Harness to Customer Interface Box Cable

STEP 2A. Check Idle/Rated Switch Signal Wire

Conditions :

- Disconnect customer interface box to engine harness cable connector C3 from the customer interface box.
- Disconnect customer interface box to engine harness cable connector C10 from the engine harness.

Action	Specification/Repair	Next Step
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Check idle/rated switch signal wire. • Place one test lead in the idle/rated switch signal pin of the C3 connector. • Place the other test lead in the idle/rated switch signal pin of the C10 connector.	Less than 10 ohms resistance? YES Repair : Refer to Section TF in the Troubleshooting and Repair Manual, QSK19 CM850 Modular Common Rail System Series Engines, Bulletin 4021493 (/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/4021493.html), or the Troubleshooting and Repair Manual, Electronic Control System, QSK38, QSK50, and QSK60, CM850 Modular Common Rail System Series Engines, Bulletin 4021533 (/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/4021533.html), or refer to the OEM service manual for idle switch repair instructions.	Repair complete.
	Less than 10 ohms resistance? NO Repair : Replace the cable.	Repair complete.

